

Lec 11-1, Nov 18, 2024

- Integration by partial fractions (8.5)

- Basic forms

- Finding undetermined coefficients

- Integration by Partial Fractions

We now introduce a method that works well for integrating rational functions $P(x)/Q(x)$. Consider finding

i.e., P and Q are polynomials $\int \frac{5x - 3}{x^2 - 2x - 3} dx.$

This becomes easy if we realize that

$$\frac{5x - 3}{x^2 - 2x - 3} = \frac{2}{x + 1} + \frac{3}{x - 3},$$

since the right-hand side is easy to integrate. How do we split $P(x)/Q(x)$ into rational functions that are easier to integrate?

$$3x^4 - 2x^2 + x - 100$$

Def. If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ with $a_n \neq 0$,
then $\deg(P(x)) = n$. Here $\deg(P(x))$ is called the degree of $P(x)$.

[We do not consider the degree of a zero polynomial here.]

e.g. $\deg(7x^3 - 2x^2 + \pi x + e) = 3$;

$$\deg(a_0) = 0 \text{ (for } a_0 \neq 0 \text{)}.$$

Fact : If $P(x)$ and $Q(x)$ are polynomials with $Q(x)$ nonzero,
then there exist polynomials $S(x)$ and $R(x)$ such that
(quotient) (remainder)

$$P(x) = Q(x) \cdot S(x) + R(x) ;$$

$$R(x) \equiv 0 \text{ or } \deg(R(x)) < \deg(Q(x)) .$$

This means that

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)} . \quad (*)$$

e.g. (Long division of polynomials).

$$\frac{x^3 - 4x^2 + 2x - 3}{x + 2} = ?$$

Sol':

$$\begin{array}{r} x^2 - 6x + 14 \\ x+2 \overline{) x^3 - 4x^2 + 2x - 3} \\ \underline{x^3 + 2x^2} \\ -6x^2 + 2x \\ \underline{-6x^2 - 12x} \\ 14x - 3 \\ \underline{14x + 28} \\ -31 \end{array}$$

$$\therefore \frac{x^3 - 4x^2 + 2x - 3}{x + 2} = x^2 - 6x + 14 + \frac{-31}{x + 2}.$$

□

By the long division of polynomials

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)},$$

we have

$$\int \frac{P(x)}{Q(x)} dx = \underbrace{\int S(x) dx}_{\textcircled{1}} + \underbrace{\int \frac{R(x)}{Q(x)} dx}_{\textcircled{2}}.$$

Since $\textcircled{1}$ is easy to integrate, we focus on $\textcircled{2}$.

Hence we may assume that

$$\deg(P(x)) < \deg(Q(x)).$$

e.g. (continued)

$$\begin{aligned} \int \frac{x^3 - 4x^2 + 2x - 3}{x+2} dx &= \int x^2 - 6x + 14 dx + \int -\frac{31}{x+2} dx \\ &= \frac{1}{3}x^3 - 3x^2 + 14x - 31 \ln|x+2| + C. \quad \square \end{aligned}$$

Factors of polynomials

e.g. $x^4 - x^2 - 2x + 2$,

It is a fact from algebra that every polynomial factors into linear or (*) irreducible quadratic polynomials over the real numbers. (A quadratic polynomial $ax^2 + bx + c$ is called irreducible if it has no real root, i.e., if $b^2 - 4ac < 0$.)

ir + reduce + able

That is, every nonzero polynomial $Q(x)$ can be written as

$$Q_1(x) Q_2(x) \cdots Q_r(x),$$

where each $Q_i(x)$ has one of the following forms:

• $a_i x + b_i$, $a_i \neq 0$; (linear)

• $a_i x^2 + b_i x + c_i$, $a_i \neq 0$, $b_i^2 - 4a_i c_i < 0$. (irreducible quadratic)
($x^2 + x + 4$)

e.g. $x^2 - 1 = ?$

Sol: $(x^2 - 1) = (x - 1)(x + 1)$

e.g. $x^2 + 1$ is irreducible in $\mathbb{R}$.

e.g. $x^4 - x^2 - 2x + 2 = ?$

Sol: $x = 1$ is a root $\Rightarrow x^4 - x^2 - 2x + 2 = (x - 1)^2 \underbrace{(x^2 + 2x + 2)}_{\text{irreducible}}.$

e.g. You can check that $x^4 + 1 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$.

↑
Rmk: This is for demonstration for the fact only.

Do not need to worry about how to factor this.

Rmk: The fact (*) is given without proof, which requires

"Fundamental Theorem of Algebra".

[over 200 different ways to prove; the first rigorous one by Gauss, 1799]

• Now we look at $\int \frac{P(x)}{Q(x)} dx$, where $\deg(P(x)) < \deg(Q(x))$.

Case 1: $Q(x)$ is a product of distinct linear factors:

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \dots (a_kx + b_k).$$

In this case, there exist constants $A_1, A_2, \dots, A_k$ such that

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}. \quad (1)$$

Example

(r) Find $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$.

Sol': Step 1. Factorize!

$$2x^3 + 3x^2 - 2x = x(2x-1)(x+2).$$

Step 2. Determine the coefficients!

$$\begin{aligned}\frac{1 \cdot x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} &= \frac{x^2 + 2x - 1}{x(2x-1)(x+2)} \\ &= \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2} \\ &= \frac{A(2x-1)(x+2) + Bx(x+2) + C \cdot x \cdot (2x-1)}{x(2x-1)(x+2)} \\ &= \frac{(2A+B+C)x^2 + (3A+2B-C)x + (-2A)}{x(2x-1)(x+2)}\end{aligned}$$

• A, B, C
are "undetermined"
coefficients.

$$\therefore \begin{cases} 2A+B+C = 1 \\ 3A+2B-C = +2 \\ -2A = -1 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{2} \\ B = \frac{1}{5} \\ C = -\frac{1}{10} \end{cases}$$

Step 3. Integrate!

$$\begin{aligned}\int \frac{P(x)}{Q(x)} dx &= \int \frac{\frac{1}{2}}{x} + \frac{\frac{1}{5}}{2x-1} + \frac{-\frac{1}{10}}{x+2} dx \\ &= \frac{1}{2} \ln|x| + \frac{1}{5} \ln|2x-1| \cdot \frac{1}{2} - \frac{1}{10} \ln|x+2| + C.\end{aligned}$$

Case 2: $Q(x)$ is a product of linear factors, some of which are repeated.

Suppose that a linear factor $(a_1x + b_1)$ is repeated r times; that is, $(a_1x + b_1)^r$ appears in the factorization of $Q(x)$. Then instead of the single term $A_1/(a_1x + b_1)$ in Equation (1) on the previous slide, we would use

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}. \quad (2)$$

Example

(r) Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx. = I.$

Sol: Step 0: Long division!

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}.$$

$\swarrow \deg(P(x))$
 $\searrow \deg(Q(x))$

Step 1: Factorize!

$$x^3 - x^2 - x + 1 = (x-1)^2(x+1).$$

Step 2: Determine the coefficients!

$$\frac{P(x)}{Q(x)} = \frac{0 \cdot x^2 + 4x + 0 \cdot 1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} = \frac{(A+C) \cdot x^2 + (B-2C)x + (-A+B+C)}{(x+1)(x-1)^2}$$

$$\Rightarrow \begin{cases} A+C=0 \\ B-2C=4 \\ -A+B+C=0 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=2 \\ C=-1 \end{cases}.$$

Step 3 : Integrate !

$$\frac{P(x)}{Q(x)} = \frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{-1}{x+1}$$

$$\therefore \int \frac{P(x)}{Q(x)} dx = \ln|x-1| + 2(x-1)^{-1} \cdot (-1) - \ln|x+1| + C.$$

Step 4 : Add !

$$I = \int (x+1) dx + \int \frac{4x}{x^3 - x^2 - x + 1} dx$$

$$= \frac{1}{2}x^2 + x + \ln|x-1| + 2(x-1)^{-1}(-1) - \ln|x+1| + C. \quad \boxed{9}.$$

Case 3: $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

If $Q(x)$ has a factor $ax^2 + bx + c$, where $b^2 - 4ac < 0$, then, in addition to the partial fractions in Equations (1) and (2), the expression for $P(x)/Q(x)$ will have a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}. \quad (3)$$

For example, there exist constants A, B, C, D and E such that

$$\frac{x}{(x-2)(x^2+1)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{x^2+4}.$$

e.g. $\int \frac{-2x+4}{(x^2+1)(x-1)^2} dx \stackrel{!}{=} I = ?$

Sol': Step 1 : Factorize !

$$Q(x) = (x^2+1)(x-1)^2.$$

[already done!]

Step 2 : Determine the coefficients !

$$\frac{0x^3 + 0x^2 - 2x + 4}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{(x-1)} + \frac{D}{(x-1)^2}$$

$$= \frac{(Ax+B)(x-1)^2 + C(x-1)(x^2+1) + D(x^2+1)}{(x^2+1)(x-1)^2}$$

$$\Rightarrow -2x+4 = \overset{0}{(A+C)}x^3 + \overset{0}{(-2A+B-C+D)}x^2 + \overset{-2}{(A-2B+C)}x + \underset{4}{(B-C+D)}$$

Solving linear equations give $A = 2, B = 1, C = -2, D = 1$.

Step 3 : Integrate!

$$I = \int \frac{2x+1}{x^2+1} dx + \int \frac{-2}{x-1} dx + \int \frac{1}{(x-1)^2} dx$$

= - - -

□

Case 4: $Q(x)$ contains a repeated irreducible quadratic factor.

If $Q(x)$ has a factor $(ax^2 + bx + c)^r$, where $b^2 - 4ac < 0$, then instead of a single term (3) on the previous slide, the sum

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

appears in the partial fraction decomposition of $P(x)/Q(x)$. For example, there exist constants $A, B, \dots, I$ and J such that

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3} = \frac{A}{x} + \frac{B}{x-1} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{x^2 + 1} + \frac{Gx + H}{(x^2 + 1)^2} + \frac{Ix + J}{(x^2 + 1)^3}$$

Integrating the Basic Forms.

After partial fraction decomposition, we need to know how to compute two types of integrals:

$$(I) \int \frac{A}{(ax+b)^n} dx ; \quad (II) \int \frac{Ax+B}{(ax^2+bx+c)^n} dx .$$

Sol^I: (I) $\int \frac{A}{(ax+b)^n} dx = \frac{A}{a} \ln |ax+b| + C .$

$$\int \frac{A}{(ax+b)^n} dx = \frac{A}{a} \frac{1}{-n+1} (ax+b)^{-n+1} + C , \text{ if } n \geq 2 .$$

For (II), we demonstrate with an example.

e.g. Compute $\int \frac{x^2+1}{(x^2-2x+2)^2} dx$ ($\triangleq I$).

Sol':

Step 1: Decompose:

$$\frac{0 \cdot x^3 + 1 \cdot x^2 + 1 + 0 \cdot x}{(x^2-2x+2)^2} = \frac{A_1 x + B_1}{x^2-2x+2} + \frac{A_2 x + B_2}{(x^2-2x+2)^2}$$

Step 2: Determine the coefficients!

$$\frac{0 \cdot x^3 + 1 \cdot x^2 + 1 + 0 \cdot x}{(x^2-2x+2)^2} = \frac{(A_1 x + B_1)(x^2-2x+2) + A_2 x + B_2}{(x^2-2x+2)^2}$$
$$= \frac{A_1 x^3 + (B_1 - 2A_1)x^2 + (-2B_1 + 2A_1 + A_2)x + B_2 + 2B_1}{(x^2-2x+2)^2}$$

$$\Rightarrow \begin{cases} A_1 = 0 \\ B_1 - 2A_1 = 1 \\ -2B_1 + 2A_1 + A_2 = 0 \\ 2B_1 + B_2 = 1 \end{cases}$$

$$\Rightarrow \begin{cases} A_1 = 0 \\ B_1 = 1 \\ A_2 = 2 \\ B_2 = -1 \end{cases}$$

$$\therefore \frac{x^2+1}{(x^2-2x+2)^2} = \frac{0 \cdot x + 1}{x^2-2x+2} + \frac{2 \cdot x + (-1)}{(x^2-2x+2)^2}$$

Step 3 : Integrate!

$$\int \frac{x^2+1}{(x^2-2x+2)^2} dx = \underbrace{\int \frac{1}{x^2-2x+2} dx}_{I_1} + \underbrace{\int \frac{2x-1}{(x^2-2x+2)^2} dx}_{I_2}.$$

$$\begin{aligned} \therefore I_1 &= \int \frac{1}{(x-1)^2+1} dx \\ &\quad \xrightarrow{y=x-1} \int \frac{1}{y^2+1} dy \Big|_{y=x-1} \\ &= \arctan y \Big|_{y=x-1} + C \\ &= \arctan (x-1) + C. \end{aligned}$$

$$\begin{aligned} &\left(\int \frac{C}{a(x-b)^2+d} dx \right) \\ &\quad \xrightarrow{y=x-b} \int \frac{C}{ay^2+d} dy \Big|_{y=x-b} \\ &= \dots \arctan \dots \end{aligned}$$

$$\therefore I_2 = \int \frac{2x-1}{(x^2-2x+2)^2} dx = \int \frac{2x-1}{((x-1)^2+1)^2} dx$$

$$\xrightarrow{y=x-1} \int \frac{2(y+1)-1}{(y^2+1)^2} dy \Big|_{y=x-1}$$

$$= \left[\int \frac{2y}{(y^2+1)^2} dy + \int \frac{1}{(y^2+1)^2} dy \right] \Big|_{y=x-1}$$

$$= \underbrace{\int \frac{2y}{(y^2+1)^2} dy}_{J} \Big|_{y=x-1} + \underbrace{\int \frac{1}{(y^2+1)^2} dy}_{K} \Big|_{y=x-1}.$$

$$\therefore J = \int \frac{1}{(y^2+1)^2} d(y^2) \Big|_{y=x-1}$$

$$\stackrel{z=y^2}{=} \int \frac{1}{(z+1)^2} dz \Big|_{z=(x-1)^2}$$

$$= -(z+1)^{-1} + C \Big|_{z=(x-1)^2}$$

$$= -\frac{1}{(x-1)^2+1} + C.$$

Step 4: Solve the hardest one by trigonometric substitutions!

$$K = \int \frac{1}{(y^2+1)^2} dy \Big|_{y=x-1}$$

$$\begin{aligned} y^2+1 &= \frac{\sin^2 \theta}{\cos^2 \theta} + 1 \\ &= \frac{1}{\cos^2 \theta} \end{aligned}$$

$$\begin{aligned} y &= \tan \theta \\ \theta &= \arctan y \end{aligned} \quad \int \frac{1}{(\tan^2 \theta + 1)^2} d(\tan \theta) \Big|_{\theta = \arctan(x-1)} \quad (\tan \theta)' = \frac{\sin' \cos - \cos' \sin}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$= \int \frac{1}{(\sec^2 \theta)^2} \sec^2 \theta d\theta \Big|_{\theta = \arctan(x-1)}$$

$$= \int \cos^2 \theta d\theta \Big|_{\theta = \arctan(x-1)}$$

$$= \int \frac{1+\cos 2\theta}{2} d\theta \Big|_{\theta = \arctan(x-1)}$$

$$= \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right] \Big|_{\theta = \arctan(x-1)}$$

$$= \frac{1}{2} \arctan(x-1) + \frac{1}{2} \frac{x-1}{1+(x-1)^2}$$

$$+ C.$$

Step 5: add all up!

$$I = I_1 + J + K$$

$$= \arctan(x-1) - \frac{1}{(x-1)^2 + 1}$$

$$+ \frac{1}{2} \arctan(x-1) + \frac{1}{4} 2(x-1) \frac{1}{1+(x-1)^2} + C.$$

$$\frac{1}{4} \sin 2\theta = \frac{1}{2} \sin \theta \cos \theta$$

$$\begin{cases} \tan \theta = x-1 \\ \sin^2 \theta + \cos^2 \theta = 1 \end{cases}$$

$$\Rightarrow \cos^2 \theta (1 + \tan^2 \theta)$$

$$= \cos^2 \theta (1 + (x-1)^2) = 1$$

$$\Rightarrow \cos^2 \theta = \frac{1}{1+(x-1)^2}$$

$$\Rightarrow \sin \theta \cos \theta$$

$$= \cos^2 \theta \cdot \tan \theta = \frac{x-1}{1+(x-1)^2}$$

$$\therefore \sin 2\theta = 2 \cdot \sin \theta \cos \theta$$

$$= 2 \cdot \frac{x-1}{1+(x-1)^2}$$

Methods of Integration

- Direct integrate, substitution: $\int h(f(x)) f'(x) dx = \int h(f(x)) df(x) \stackrel{y=f(x)}{=} \int h(y) dy$
- Integration by parts: $\int f(x) g'(x) dx = \int f(x) dg(x) = f(x)g(x) - \int g(x) df(x)$
- Trigonometric integration (reduction): $\int \sin^m x \cos^n x dx$, $\int \tan^m x dx$, $\int \sec^n x dx$,
 $\left[\begin{array}{l} \tan x \leftrightarrow \sec x \\ \sin x \leftrightarrow \cos x \end{array} \right] \quad \int \tan^m x \sec^n x dx$
- Trigonometric substitution: $\int \frac{1}{\sqrt{1-x^2}} dx$, $\int \frac{1}{\sqrt{x^2+1}} dx$, $\int \frac{1}{(x^2-1)^{1/2}} dx$ $\left[\begin{array}{l} x = \tan \theta \\ \sec \theta \\ \sin \theta \end{array} \right]$
- Integration by partial fractions: $\int \frac{P(x)}{Q(x)} dx$
 $\underbrace{\hspace{1cm}} \Rightarrow \int S(x) + \frac{R(x)}{Q(x)} dx$

· Finding undetermined coefficients : Heaviside "cover up" method

In case 1, where

$$(x-r_1) \cdot \frac{f(x)}{g(x)} = \frac{f(x)}{(x-r_1) \cdots (x-r_n)} \cdot (x-r_1) \\ = \frac{A_1}{\cancel{x-r_1}} + \left(\cdots + \frac{A_n}{x-r_n} \right) \cdot (x-r_1)$$

there is a quick way for finding $A_1, \dots, A_n$.

Indeed, by multiplying both sides by $x-r_1$, we have

$$\left. \frac{f(x)}{(x-r_2) \cdots (x-r_n)} \right|_{x=r_1} = A_1 + \left(\frac{A_2}{\cancel{(x-r_2)}} + \cdots + \frac{A_n}{x-r_n} \right) \cdot \left. (x-r_1) \right|_{x=r_1}$$

Substituting $x=r_1$ yields

$$A_1 = \left. \frac{f(x)}{(x-r_2) \cdots (x-r_n)} \right|_{x=r_1} = \frac{f(r_1)}{(r_1-r_2) \cdots (r_1-r_n)}$$

["cover up" $\underbrace{(x-r_1)}$ in $\frac{f(x)}{g(x)}$, then sub in $x=r_1$: hence the name].

The Similar approach works for all the A_i 's: remove $x - r_i$ from the bottom of $\frac{f(x)}{g(x)}$, then sub in $\underline{x = r_i}$ to get A_i .

e.g. In one of our previous examples, we have

$$\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} = \frac{f(x)}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2}.$$

"Cover up"!

$$A = \left. \frac{f(x)}{(2x-1)(x+2)} \right|_{x=0} = \left. \frac{x^2 + 2x - 1}{(2x-1)(x+2)} \right|_{x=0} = \frac{1}{2}$$

$$B = \left. \frac{f(x)}{x(x+2)} \right|_{x=\frac{1}{2}} = \left. \frac{x^2 + 2x - 1}{x(x+2)} \right|_{x=\frac{1}{2}} = \frac{\frac{1}{4} + 1 - 1}{\frac{1}{2} \cdot \frac{5}{2}} = \frac{1}{5}.$$

$$C = \left. \frac{f(x)}{x(2x-1)} \right|_{x=-2} = -\frac{1}{10}.$$

This approach may be modified slightly to handle other cases.

e.g. $\frac{3x^2 - 16x + 21}{(x-1)^2(x+3)} = \frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{C}{(x+3)}.$

Sol': Step 1: Cover up the highest degree!

$$(x-1)^2 \frac{3x^2 - 16x + 21}{(x-1)^2 (x+3)} = (x-1)^2 \left[\frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{C}{(x+3)} \right]$$

• To get A, cover up (x-1) on the LHS and evaluate at $x=1$:

$$A = \left. \frac{3x^2 - 16x + 21}{x+3} \right|_{x=1} = \frac{3 - 16 + 21}{4} = 2$$

Step 2 : move it to the LHS !

$$\begin{aligned} \frac{3x^2 - 16x + 21}{(x-1)^2 (x+3)} - \frac{2}{(x-1)^2} &= \frac{(3x^2 - 16x + 21) - 2(x+3)}{(x-1)^2 (x+3)} \\ &= \frac{3x^2 - 18x + 15}{(x-1)^2 (x+3)} \end{aligned}$$

$$\Rightarrow \frac{3x^2 - 18x + 15}{(x-1)^2 (x+3)} = \frac{3x - 15}{(x-1)(x+3)} = \frac{B}{x-1} + \frac{C}{x+3}$$

Step 3 : Cover up the rest !

Now B and C can be found by the cover-up method again.

$$\cancel{(x-1)} \frac{3x - 15}{\cancel{(x-1)}(x+3)} \bigg|_{x=1} = \cancel{(x-1)} \left[\frac{B}{\cancel{x-1}} + \frac{C}{x+3} \right] \bigg|_{x=1} \Rightarrow B = \left. \frac{3x - 15}{x+3} \right|_{x=1} = -3$$

Integration by Partial Fractions

Finding undetermined coefficients : differentiating

This technique will be useful for functions of the form $\frac{f(x)}{(x-r)^n}$.

e.g. $\frac{f(x)}{(x-r)^4} = \frac{A}{(x-r)^4} + \frac{B}{(x-r)^3} + \frac{C}{(x-r)^2} + \frac{D}{x-r}$.

What are A, B, C, D?

Sol': $\frac{f(x)}{(x-r)^4} = \frac{A + B(x-r) + C(x-r)^2 + D(x-r)^3}{(x-r)^4}$

1. $f(x) = A + B(x-r) + C(x-r)^2 + D(x-r)^3$. (1)

$\Rightarrow f(r) = A$.

2. (1) $\Rightarrow f'(x) = 0 + B + 2C(x-r) + D \cdot 3(x-r)^2$. (2)

$\Rightarrow f'(r) = B$.

3. (2) $\Rightarrow f''(x) = 0 + 0 + C \cdot 2 + D \cdot 3 \cdot 2(x-r)$. (3)

$\Rightarrow f''(r) = 2C \quad \Rightarrow C = \frac{1}{2} f''(r)$.

4. (3) $\Rightarrow f'''(x) = 0 + 0 + 0 + D \cdot 3 \cdot 2 \cdot 1$

$\Rightarrow f'''(r) = 6D \quad \Rightarrow D = \frac{1}{6} f'''(r)$.

e.g. Find the partial fraction decomposition for $\int \frac{3x^2 - 16x + 21}{(x-1)^2(x+3)} dx$

Sol¹: Step 1: Decompose!

$$\frac{3x^2 - 16x + 21}{(x-1)^2(x+3)} = \frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{C}{x+3}.$$

Step 2: Use Heaviside's cover-up method to get C!

$$\cancel{(x+3)} \frac{3x^2 - 16x + 21}{(x-1)^2 \cancel{(x+3)}} \Big|_{x=-3} = C + \left(\frac{A}{(x-1)^2} + \frac{B}{(x-1)} \right) \cancel{(x+3)} \Big|_{x=-3}$$

$$\Rightarrow C = \frac{3x^2 - 16x + 21}{(x-1)^2} \Big|_{x=-3} = \frac{48 + 48}{16} = 6.$$

Step 3: Subtract $\frac{C}{x+3}$!

$$\frac{3x^2 - 16x + 21}{(x-1)^2(x+3)} - \frac{6}{x+3} = \frac{-3x + 5}{(x-1)^2} = \frac{A}{(x-1)^2} + \frac{B}{(x-1)}$$

Step 4 : Use differentiation method !

$$\frac{f(x)}{(x-1)^2} = \frac{A}{(x-1)^2} + \frac{B}{x-1}$$

$$\Rightarrow \begin{cases} f(x) = A + B(x-1) \\ f'(x) = B \end{cases}$$

$$\Rightarrow \begin{cases} f(1) = A = (-3x+5)|_{x=1} = 2 \\ f'(1) = B = (-3x+5)' = -3 \end{cases}$$

$$\therefore \frac{3x^2-16x+21}{(x-1)^2(x+3)} = \frac{2}{(x-1)^2} + \frac{-3}{(x-1)} + \frac{6}{(x+3)}$$

□ .